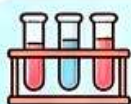


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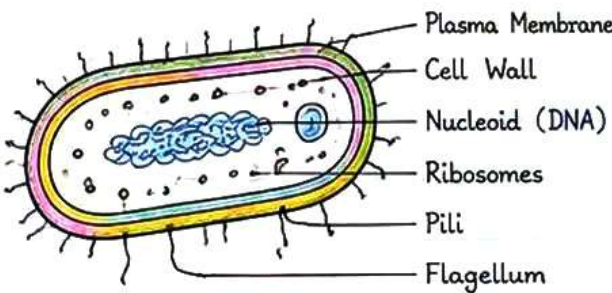
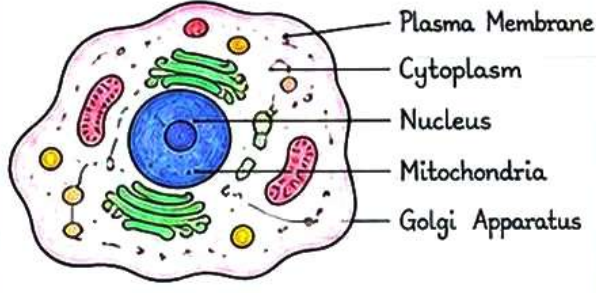
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* 1. Types of Cells

- On the basis of presence or absence of a true nucleus and membrane-bound organelles, cells are of two types:

- Prokaryotic Cells
- Eukaryotic Cells

A. Prokaryotic vs Eukaryotic Cells

Prokaryotic Cells	Eukaryotic Cells
- Small in size (1-10 μm). - No true nucleus. - Nuclear material is naked and lies in a region called nucleoid. - Membrane-bound organelles are absent. - Circular DNA present. 70S ribosomes are present. - Cell wall is present (made of peptidoglycan). - Examples - Bacteria, Blue-green algae (Cyanobacteria), Mycoplasma.	- Larger in size (10-100 μm). - True nucleus is present. - Nuclear material is enclosed by a nuclear membrane. - Membrane-bound organelles are present. - Linear DNA present. 80S ribosomes are present. - Cell wall is absent in animals, present in plants (made of cellulose). - Examples - Amoeba, Paramecium, Plant cells, Human cells.
 (A Typical Prokaryotic Cell)	 (A Typical Eukaryotic Cell)

B. Cell Size & Shape

- Cells vary greatly in their size and shape.
- Size - Generally measured in micrometre (μm).
 - Prokaryotic cells: 1-10 μm
 - Eukaryotic cells: 10-100 μm
- Shape - Cells show a variety of shapes depending on their function.

→ Spherical - e.g., Ovum	}	→ Irregular - e.g., Amoeba
→ Rod-shaped - e.g., Bacteria		→ Disk-shaped - e.g., RBC (Red Blood Cell)
→ Elongated - e.g., Muscle cells		

* Cell size and shape are related to their specific functions in an organism.

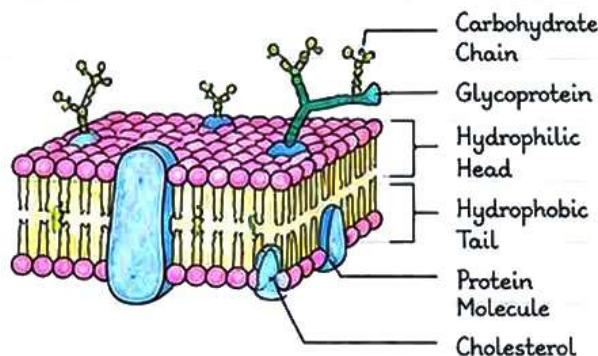


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* 3. Plasma Membrane

- Plasma membrane (Cell membrane) is a thin, living, selectively permeable membrane that surrounds the cytoplasm of the cell.
- It separates the cell from its external environment and controls the movement of substances in and out of the cell.



(Structure of Plasma Membrane)

A. Structure

- It is a unit membrane, 7-10 nm thick.
- It is composed of lipids (phospholipids, cholesterol), proteins and carbohydrates.
- It shows trilaminar appearance under electron microscope.

B. Fluid Mosaic Model (S. J. Singer and G. L. Nicolson, 1972)

- According to this model, the plasma membrane is a dynamic structure.
- Phospholipids are arranged in a bilayer with hydrophilic heads facing outward and hydrophobic tails facing inward.
- Proteins are embedded in the lipid bilayer like pieces in a mosaic.
- Carbohydrates are present on the outer surface in the form of glycoproteins and glycolipids.
- The membrane is fluid as lipids and some proteins can move laterally.

Fluid + Mosaic
= Fluid Mosaic
Model

C. Functions

- It separates the cell from the external environment.
- It protects the cell from mechanical injury and infection.
- It controls the entry and exit of substances.
- It helps in cell recognition and cell-to-cell communication.
- It maintains the shape of the cell.
- It acts as a receptor site for various molecules like hormones.
- It helps in endocytosis and exocytosis.
- It maintains ionic balance and membrane potential.

D. Transport Across Plasma Membrane

① Diffusion

- Movement of molecules from a region of higher concentration to a region of lower concentration.
- It does not require energy.
- Example - Diffusion of O_2 , CO_2 ,

High Concentration → Low Concentration

② Osmosis

- Movement of water molecules from a region of higher water potential (dilute solution) to a region of lower water potential (concentrated solution) through a semipermeable membrane.

High Water Potential → Low Water Potential

③ Active Transport

- Movement of molecules from a region of lower concentration to a region of higher concentration with the help of energy (ATP).
- It occurs against the concentration gradient.
- Example - Uptake of ions like Na^+ , K^+ .

Low Concentration → High Concentration (with energy)

* Plasma membrane is selectively permeable and plays a vital role in maintaining the internal environment of the cell.

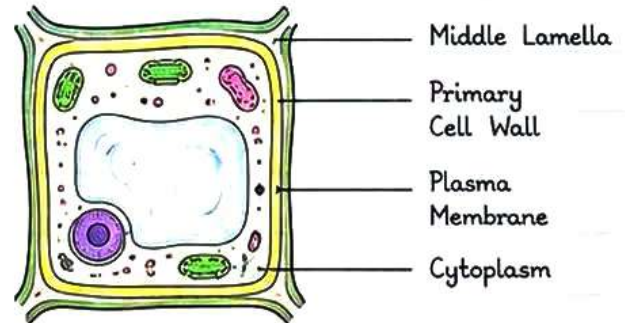


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* 4. Cell Wall

- Cell wall is a rigid, non-living covering present outside the plasma membrane in plant cells, fungi, algae and bacteria.
- It provides shape, strength and protection to the cell.
- It is freely permeable to water and solutes.



(Position of Cell Wall in Plant Cell)

A. Structure

- It is composed mainly of cellulose, hemicellulose, pectins and proteins.
- Cellulose microfibrils provide strength and rigidity.
- Pectin is a complex polysaccharide that cements the cells together.
- It is a non-living, extracellular structure.

B. Functions

- Provides definite shape and mechanical strength.
- Protects the cell from mechanical injury.
- Prevents osmotic bursting of the cell.
- Freely permeable, allows the movement of water and solutes.
- Helps in cell-to-cell interaction and adhesion.
- Provides support for the plant body.

C. Middle Lamella

- The middle lamella is the outermost layer of the cell wall.
- It is present between two adjacent cells.
- It is made up of calcium and magnesium pectate.
- It cements the cells together, thus providing cohesion and strength to the tissues.

D. Primary and Secondary Wall

1. Primary Wall

- It is the first wall formed during cell growth.
- It is thin, flexible and extensible.
- Made up of cellulose, hemicellulose, pectins and proteins.
- Allows cell to grow and increase in size.
- Present in young and actively growing cells.

2. Secondary Wall

- It is formed after the primary wall.
- It is thick, rigid and non-extensible.
- Mainly made up of cellulose.
- Provides additional strength and protection.
- Present in mature and non-growing cells.

Primary Wall	Secondary Wall
- First wall formed. - Thin, flexible and extensible. - Made up of cellulose, hemicellulose, pectins and proteins. - Allows cell to grow. - Present in young and growing cells.	- Formed after primary wall. - Thick, rigid and non-extensible. - Mainly made up of cellulose. - Provides strength and protection. - Present in mature and non-growing cells.

- * Cell wall is a unique feature of plant cells and plays a crucial role in maintaining their shape, strength and survival.



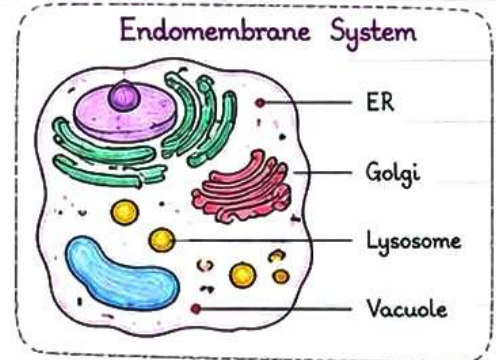
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* 5. Endomembrane System

The endomembrane system is a group of membranes that work together in the cell for synthesis, modification, packaging and transport of materials.

→ It includes : Endoplasmic Reticulum, Golgi Apparatus, Lysosomes, Vacuoles and associated vesicles.

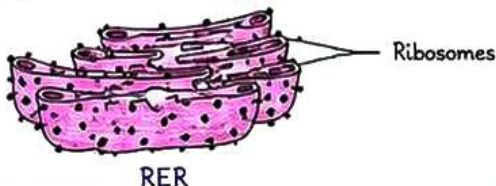


A. Endoplasmic Reticulum (ER)

- ER is a network of membranous tubules, vesicles and flattened sacs (cisternae).
- It is continuous with the nuclear envelope.

1. Rough Endoplasmic Reticulum (RER)

- Ribosomes are attached on its surface.
- Involved in protein synthesis.
- Proteins are modified and transported to Golgi apparatus.
- Abundant in cells secreting proteins (e.g., pancreas, salivary glands).



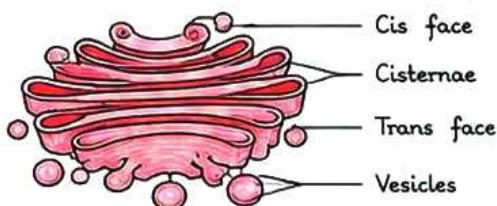
2. Smooth Endoplasmic Reticulum (SER)

- Ribosomes are absent on its surface.
- Involved in synthesis of lipids, steroids and phospholipids.
- Helps in detoxification of drugs and poisons.
- Helps in carbohydrate metabolism.
- Forms sarcoplasmic reticulum in muscle cells (stores Ca^{2+} ions).



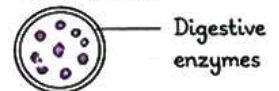
B. Golgi Apparatus

- Discovered by Camillo Golgi.
- Composed of a stack of flattened, membrane-bound sacs called cisternae with small vesicles around.
- Has two faces : cis face (forming face) and trans face (maturing face).
- Functions :
 - Modification of proteins and lipids.
 - Packaging and sorting of materials.
 - Formation of secretory vesicles and lysosomes.
 - Secretion of cell products.



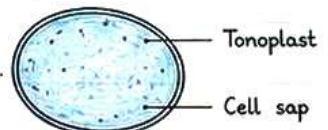
C. Lysosomes

- Small, spherical, single-membrane bound bodies containing digestive enzymes.
- Enzymes remain inactive in cytoplasm and become active in acidic medium inside lysosomes.
- Functions :
 - Intracellular digestion of food particles.
 - Removal of worn-out organelles (autophagy).
 - Digest foreign particles (defence mechanism).



D. Vacuoles

- Membrane-bound sacs filled with cell sap (water, salts, sugars, pigments, wastes).
- Large central vacuole is present in plant cells.
- Functions :
 - Storage of food, water and wastes.
 - Maintains turgor pressure and rigidity.
 - Helps in growth of the cell.



* Endomembrane system works together to maintain cell organization and transport materials efficiently within the cell.

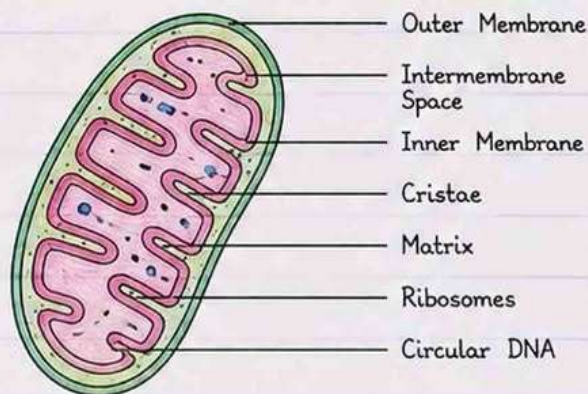


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* 6. Mitochondria

- Mitochondria are double membrane-bound organelles found in almost all eukaryotic cells.
- They are known as "Powerhouses of the cell" because they produce energy in the form of ATP.
- They vary in shape (rod-shaped, oval or spherical) and size ($0.5 - 1.0 \mu\text{m}$ in diameter and $1 - 10 \mu\text{m}$ in length).



(Structure of Mitochondria)

A. Structure

- It is surrounded by a double membrane - outer membrane is smooth and inner membrane is highly folded to form cristae.
- The space between the two membranes is called intermembrane space.
- The inner compartment enclosed by the inner membrane is called matrix.
- Matrix contains enzymes, ribosomes (70S), circular DNA, RNA and granules.
- F_1 particles (elementary particles) are present on the inner membrane (on the surface of cristae) which are involved in ATP synthesis.

B. Functions

- ① Production of ATP (energy) through aerobic respiration (Krebs cycle and Electron Transport Chain).
- ② Site for aerobic oxidation of carbohydrates, fats and proteins.
- ③ Involved in regulation of calcium ions (Ca^{2+}) in the cell.
- ④ Help in heat production in brown fat.
- ⑤ Involved in apoptosis (programmed cell death).
- ⑥ Semi-autonomous - has its own DNA and ribosomes.

C. Endosymbiotic Theory

- Proposed by Lynn Margulis in 1967.
- According to this theory, mitochondria originated from free-living aerobic bacteria that entered a host cell.
- The host cell did not digest these bacteria and instead formed a mutualistic relationship.
- Over millions of years, these bacteria lost their independent life and became mitochondria.
- Evidences in support:
 - ① Presence of circular DNA similar to bacteria.
 - ② 70S ribosomes like in prokaryotes.
 - ③ Double membrane structure.
 - ④ Ability to divide independently by binary fission.
 - ⑤ Similar in size to bacteria.

* Mitochondria are the sites of cellular respiration and ATP production, playing a vital role in energy metabolism and survival of the cell.



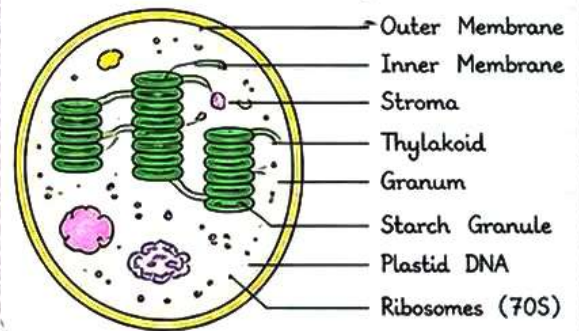
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* 7. Plastids

- Plastids are membrane-bound organelles present in plant cells and algal cells.
- They contain their own DNA, 70S ribosomes and can divide independently.
- They are of three types based on pigments present: Chloroplasts, Chromoplasts and Leucoplasts.

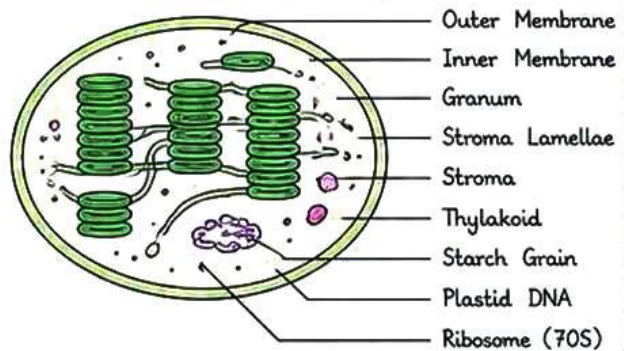
General Structure of Plastid



A. Chloroplast

- Chloroplasts are green coloured plastids due to the presence of chlorophyll.
- Main site of photosynthesis.
- Present in green parts of plants like leaves and green stems.
- Each chloroplast has a double membrane, grana (stacks of thylakoids) and stroma.
- Thylakoids contain chlorophyll and are the site of light reaction.
- Stroma contains enzymes for dark reaction (Calvin cycle).

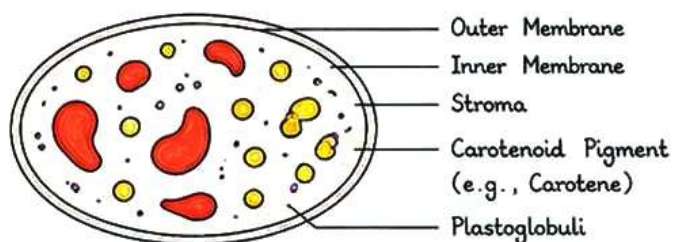
Structure of Chloroplast



B. Chromoplast

- Chromoplasts are coloured plastids.
- They contain pigments other than chlorophyll.
- Pigments include carotenoids (carotenes - orange, and xanthophylls - yellow).
- They impart colour to fruits, flowers and leaves.
- They do not take part in photosynthesis.

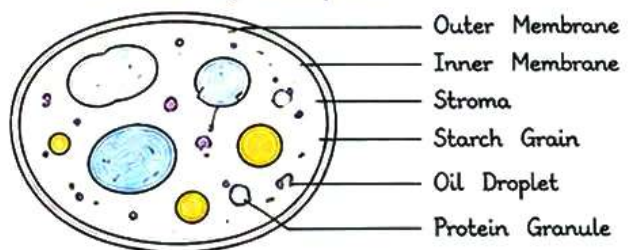
Structure of Chromoplast



C. Leucoplast

- Leucoplasts are colourless plastids.
- They lack pigments.
- They are found in non-green parts of plants like roots, tubers, seeds and endosperm.
- They store food materials in the form of starch, oils or proteins.

Structure of Leucoplast



* Plastids are semi-autonomous organelles having their own DNA and 70S ribosomes. They can convert from one type to another depending on the need.



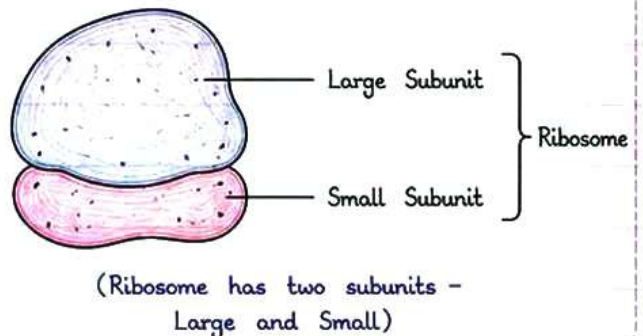
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* 8. Ribosomes

- Ribosomes are small, spherical, non-membranous structures present in all living cells.
- They are the sites of protein synthesis.
- They are made up of ribosomal RNA (rRNA) and proteins.
- They may occur freely in the cytoplasm or be attached to the endoplasmic reticulum.

Structure of Ribosome



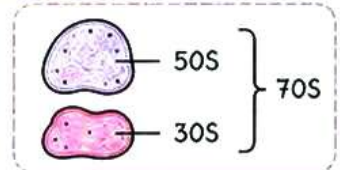
A. Structure

- Each ribosome consists of two subunits: a large subunit and a small subunit.
- These two subunits are made up of rRNA and proteins.
- During protein synthesis, both subunits come together to form a functional ribosome.
- After the completion of protein synthesis, the subunits separate again.

B. Types of Ribosomes

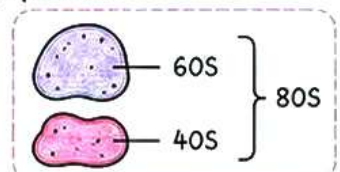
1. 70S Ribosomes

- Found in prokaryotic cells (bacteria and cyanobacteria) and in organelles like mitochondria and plastids.
- Composed of:
 - Large subunit - 50S
 - Small subunit - 30S
 - Total - 70S



2. 80S Ribosomes

- Found in eukaryotic cells (cytoplasm).
- Composed of:
 - Large subunit - 60S
 - Small subunit - 40S
 - Total - 80S



C. Functions

- ① Protein Synthesis : Ribosomes are the site of protein synthesis. They translate the genetic information from mRNA into polypeptide chains.
- ② Growth and Repair : Proteins formed by ribosomes are essential for growth, repair and maintenance of cells.
- ③ Enzyme Formation : Many enzymes are proteins synthesized on ribosomes.
- ④ Structural Role : Ribosomes help in the formation of structural proteins required for cell structure and function.
- ⑤ Essential for Life : Since they are involved in protein synthesis, ribosomes are essential for the survival of all living organisms.

* Ribosomes are tiny but vital organelles responsible for protein synthesis, which is essential for all cellular activities and life processes.



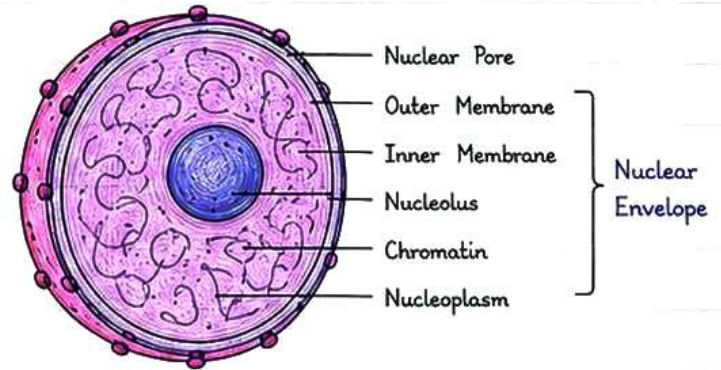
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* 9. Nucleus

- Nucleus is a large, spherical or oval, double membrane-bound organelle.
- It is the control centre of the cell.
- It contains genetic material (DNA) in the form of chromatin.
- It plays an important role in heredity, growth, metabolism and cell division.

Structure of Nucleus

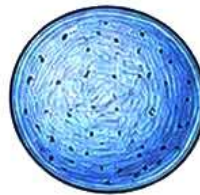


A. Nuclear Envelope (or Membrane)

- Nucleus is surrounded by a double membrane called nuclear envelope.
- It consists of an outer membrane and an inner membrane separated by a narrow perinuclear space.
- The outer membrane is continuous with the endoplasmic reticulum.
- The envelope is perforated by nuclear pores which regulate the movement of substances between nucleus and cytoplasm.
- It protects the contents of the nucleus and maintains its shape.

B. Nucleolus

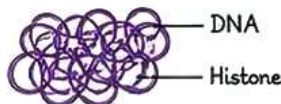
- Nucleolus is a small, dense, spherical structure present inside the nucleus.
- It is not membrane-bound.
- It is the site of rRNA synthesis and ribosome assembly.
- During cell division, nucleolus disappears and reappears after division.



- Non-membranous
- Dense and spherical
- Site of rRNA synthesis and ribosome assembly

C. Chromatin

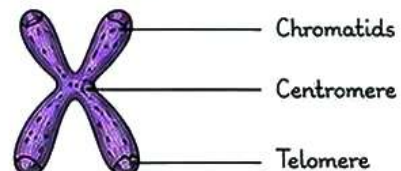
- Chromatin is a network of long, thin threads made up of DNA and proteins (histones).
- It is present in the nucleoplasm.
- During cell division, chromatin condenses and becomes visible as chromosomes.
- It carries genetic information and controls the activities of the cell.



- * Nucleus is the command centre of the cell. It stores genetic information and regulates all cellular activities.

D. Chromosome

- Chromosomes are thread-like structures formed by the condensation of chromatin during cell division.
- Each chromosome is made of DNA and proteins.
- They carry genes which determine the characters and traits.
- In a species, the number of chromosomes is constant.
- They are of different shapes - rod-shaped, V-shaped, J-shaped, etc.





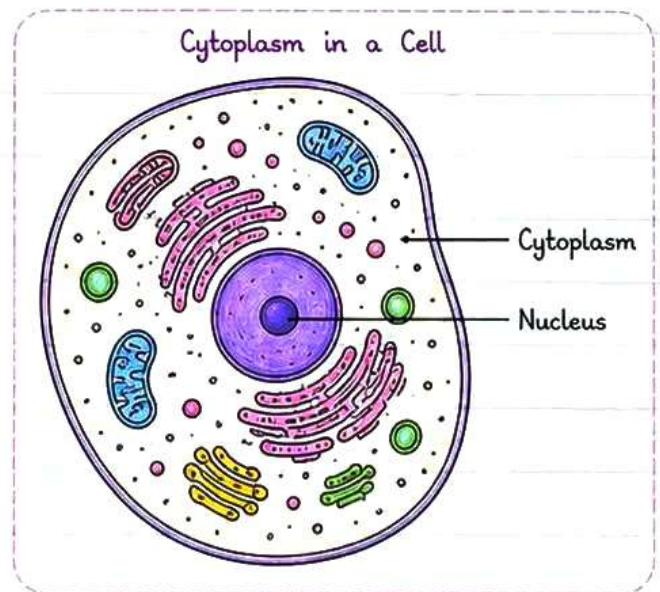
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* 10. Cytoplasm + Cytoskeleton

A. Cytoplasm

- Cytoplasm is the jelly-like substance present between the plasma membrane and the nucleus.
- It includes cytosol (matrix) and all the organelles present in it.
- Cytosol is a semi-fluid, colourless substance composed of water, ions, enzymes and proteins.
- It is the site of many metabolic reactions such as glycolysis.
- It helps in the movement of organelles and intracellular transport.
- It maintains cell shape and supports cell organelles.



B. Cytoskeleton

- Cytoskeleton is a network of protein filaments present in the cytoplasm.
- It maintains cell shape, provides mechanical strength and is involved in movement.
- It is of three types : Microtubules, Microfilaments and Intermediate Filaments.

① Microtubules

- These are hollow, cylindrical structures.
- Made up of tubulin protein.
- Diameter : ~25 nm
- Functions :
 - ① Maintain cell shape.
 - ② Form spindle fibres during cell division.
 - ③ Help in intracellular transport.
 - ④ Form cilia and flagella.



(Structure of Microtubule)

② Microfilaments

- These are thin, solid filaments.
- Made up of actin protein.
- Diameter : ~7 nm
- Functions :
 - ① Help in cell movement.
 - ② Involved in muscle contraction.
 - ③ Help in cytokinesis (cleavage furrow formation).
 - ④ Maintain cell shape and support.



(Structure of Microfilament)

③ Intermediate Filaments

- These are rope-like, fibrous proteins.
- Made up of various proteins like keratin, vimentin, desmin, etc.
- Diameter : ~10 nm
- Functions :
 - ① Provide mechanical strength.
 - ② Resist mechanical stress.
 - ③ Maintain cell and nuclear integrity.



(Structure of Intermediate Filament)

* Cytoplasm is the living ground of the cell where all organelles perform their functions, and cytoskeleton acts as the cell's internal framework.



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* 11. Cilia & Flagella

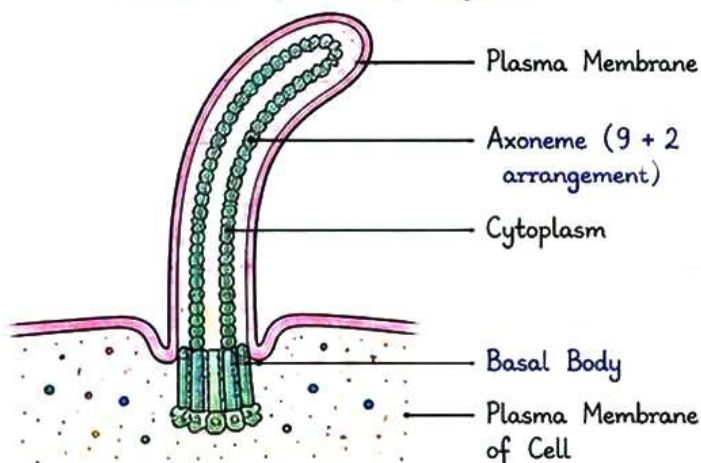
A. Structure

- Cilia and flagella are long, slender, hair-like projections present on the surface of some eukaryotic cells.
- Both are covered by plasma membrane and contain a core structure called axoneme.
- The axoneme shows a 9 + 2 arrangement of microtubules.
- There are nine peripheral doublets of microtubules arranged in a circle around two central microtubules.
- The doublets are connected by nexin links and have dynein arms which help in movement.

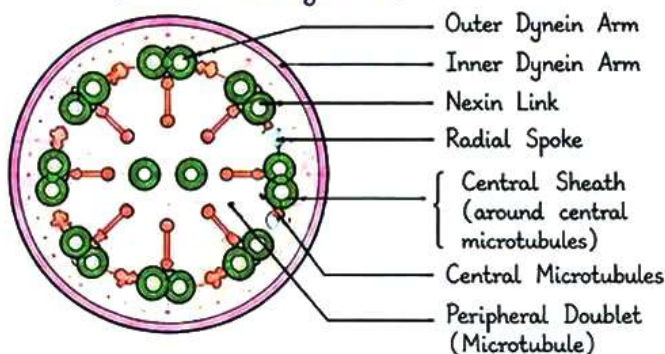
B. Functions

- Cilia and flagella are responsible for movement.
- Cilia are short and numerous; they move in a coordinated manner to move the cell or move substances over the cell surface.
- Flagella are long and few in number; they help in locomotion of the cell.
- They also help in creating currents in fluids.
- Example - Movement of mucus in respiratory tract by cilia; movement of sperm by flagellum.

Structure of Cilium / Flagellum

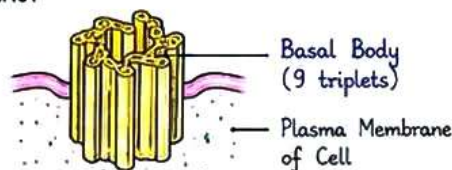


Cross section of Axoneme (9 + 2 Arrangement)



C. Basal Body

- Each cilium or flagellum arises from a basal body present just beneath the plasma membrane.
- Basal body is cylindrical and is made up of nine triplets of microtubules arranged in a ring (9 × 3 pattern).
- It lies perpendicular to the axoneme.
- It acts as the organizing centre for the formation of axoneme during development.
- It anchors the cilium or flagellum to the cell.



* Cilia and flagella show 9 + 2 arrangement of microtubules in their axoneme. They are involved in movement of cells and substances and arise from basal bodies.